.2 evidence of collision could only be found for the period 2005 - 2017 for the sperm whale in the Mediterranean;

.3 the majority of fin whales fatally struck by ships had not yet reached the reproductive stage; and

.4 the small size of the fin whale population in Mediterranean waters makes it particularly vulnerable to anthropogenic pressures.

3.5 An assessment carried out by France as part of the implementation of the European Marine Strategy Framework Directive (MSFD), in 2018, reports that in the western

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Mediterranean collisions are a cause for concern for fin whales, accounting for 80% of recorded events, compared to 10% for sperm whales (Spitz et al., 2018). Other work indicates that collisions and incidental catches alone may be responsible for the decline of the Mediterranean fin whale subpopulation, and points to the need for further research to determine how indirect anthropogenic mortalities (pollution, prey depletion) affect the sperm whale population (Sèbe et al., 2020).

3.6 Another approach to assessing the risk of collision is theoretical statistical analysis. Thus, the processing of data concerning shipping traffic with those mentioning the presence

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3.6 Another approach to assessing the risk of collision is theoretical statistical analysis. Thus, the processing of data concerning shipping traffic with those mentioning the presence
of cetaceans made it possible to calculate a theoretical ship-whale encounter rate ("near miss event" or NME). This approach was implemented for the study area (excluding the Spanish corridor), and gives the following results for fin whales (Gallou and Folegot, 2020), however this work could not be carried out for sperm whales due to the lack of sufficient biological data:

.1 seasonal differences are mainly due to the variability in the number of ships using the area, which doubles in summer compared to winter; and

.2 passenger ships and cargo ships have the highest cumulative risk of collision (84% NME in winter, 72% in summer).

3.7 In a similar manner the Spanish experience focused on a study carried out in the MPA Cetacean Migration Corridor in the Mediterranean (CEDEX, 2021), where the presence of fin whales) and sperm whales has been confirmed. For this purpose, a spatial qualitative indicator

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Cetacean Migration Corridor in the Mediterranean (CEDEX, 2021), where the presence of fin whales) and sperm whales has been confirmed. For this purpose, a spatial qualitative indicator of "potential risk of collision" was used considering, on the one hand, data related to maritime traffic, based on AIS data, and on the other hand, the available information related to sightings of the species under study, cited above.

3.8 The analysis carried out for the period of Oct 2018-Sep 2019 showed that up to 4,552 ships (including high-speed crafts, passenger ships, cargoes and tankers) have transited this marine protected area, making a total of 5 .81 million km travelled with an average route per ship of 132 km.

3.9 In order to obtain the spatial distribution of collision risk, a hazard analysis was carried out, based on the logistic curve that relates the ship's speed and mortality (Vanderlaan and Taggart, 2007) and an approximation to a hazard index based on Vaes and Druon (2013). This index includes not only the traffic involved, but also the characteristics of the ship and its

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Taggart, 2007) and an approximation to a hazard index based on Vaes and Druon (2013). This index includes not only the traffic involved, but also the characteristics of the ship and its navigation features (i.e. distance travelled), which can affect the fate of the cetacean after the collision. This concept of risk represents a further step, since it combines the hazard of maritime traffic with the exposure associated with the presence of cetaceans.

3.10 The final objective was to identify those zones within the study area where the concentration of individuals and overall risk was higher. Within the cetacean migration corridor, for the total traffic analysed, these areas were identified with the nort h-western end of the corridor and the area affected by the routes starting from the port of Barcelona, as shown in figure 7. A more detailed analysis (not included in this document) makes it possible to quantify the contribution to this risk indicator of the different categories of ships or the incidence of the seasonal effects of traffic.

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the contribution to this risk indicator of the different categories of ships or the incidence of the seasonal effects of traffic.

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Balaenoptera physalus Physeter macrocephalus

Figure 7: Potential collision risk index associated with the presence of the cetaceans and maritime traffic in the CCM Oct 2018-Sept 2019 (CEDEX, 2021).

3.11 In order to help the decision-making process, this analysis allows focusing on where and when to take measures (i.e. depending on the availability of data related to a temporal distribution of cetaceans). Based on the above, it is concluded that the whale population has suffered ship strikes in the region and therefore the cetacean population is at risk. Without associated protective measures to mitigate the risk of collision within the perimeter of the PSSA, a decline in the populations of medium and large cetaceans is to be expected.

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associated protective measures to mitigate the risk of collision within the perimeter of the PSSA, a decline in the populations of medium and large cetaceans is to be expected. Implementing a speed reduction strategy will allow a significant decrease in the likeli hood of collision and fatal wildlife-related injuries.

3.12 The IWC Scientific Committee has identified the need for a better understanding of the relationship between vessel speed, the risk of death or injury to the whale and damage to the vessel. It has considered a number of studies and approaches since 200 9 when MEPC.1/Circ.674 was adopted. All the studies considered have confirmed an increased risk with increased speed, supporting the use of speed restrictions as a way of reducing risk. Some studies have attempted to quantify the speed-risk relationship for specific whale species (Conn and Silber, 2013) or the hydrodynamic forces in relation to speed (Silber et al., 2014). Others (e.g. Wiley et al., 2011) have evaluated the relative risk reduction that might be achieved by speed restrictions. In addition to studies based on collisions, studies based on

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Others (e.g. Wiley et al., 2011) have evaluated the relative risk reduction that might be achieved by speed restrictions. In addition to studies based on collisions, studies based on observations of whales close to vessels have inferred greater collision risks with increases in speed (Gende et al., 2011; Harris et al., 2012).

3.13 At its meeting in 2022, the IWC Scientific Committee recommended that "action needs to be taken to reduce ship strike risks to the Mediterranean populations of fin and sperm whales". The Committee also recognized that, "in line with its previous recommendations, since routeing options do not seem to be possible in the area, the most effective way to reduce risk is through speed reductions ". Finally, the Committee recommended that "any measures that are implemented are fully monitored and evaluated in t erms of the risk reduction that is expected to be achieved, including through the use of AIS data to assess levels of industry cooperation, and that measures can be adapted based on this".

3.14 The most recent example of voluntary speed reduction to mitigate cetacean ship

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expected to be achieved, including through the use of AIS data to assess levels of industry cooperation, and that measures can be adapted based on this".

3.14 The most recent example of voluntary speed reduction to mitigate cetacean ship strikes is given by the case of the endangered Bryde 's whales in the Hauraki Gulf, New Zealand (Constantine et al., 2015). Since the introduction of a speed limit of 10 knots in 2013, no collision events were recorded after an average of 2.4 whales per annum recorded in the period 1996 - 2014 (Ebdon et al., 2019).
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3.15 Along the Atlantic coast of the United States, in the five years after the enactment of mandatory 10 knots speed restrictions in several Seasonal Management Areas, there were no right whale mortalities attributed to ship strikes either in or within 45 NM o f these areas. These results indicate a statistically significant reduction in right whale ship lethal strikes in

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